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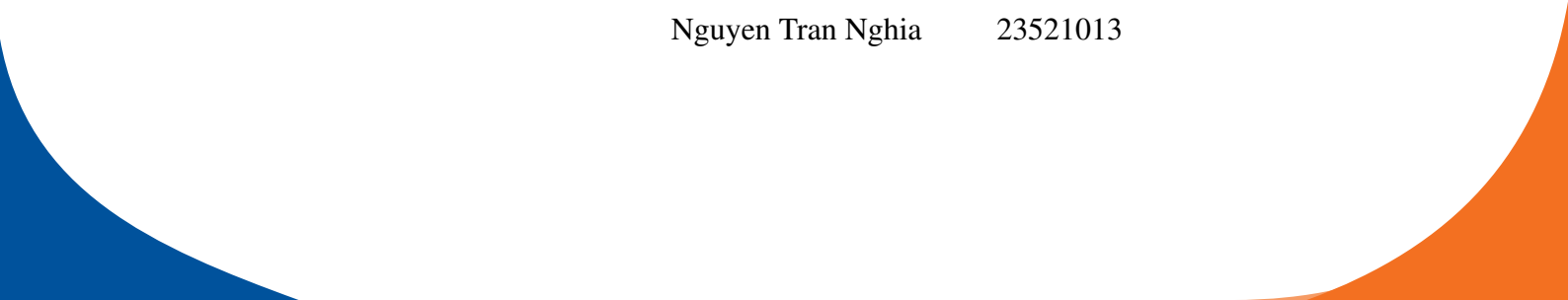
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**PROJECT REPORT
CS313.Q23 — MACHINE LEARNING**

TOPIC
Social Anxiety Detection for Humanity

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1 INTRODUCTION

In today's fast-paced world, chronic stress has become a silent epidemic, heavily impacting our mental and physical well-being. To address this growing challenge, our project introduces a proactive, machine learning-based system designed to predict individual stress levels before they become overwhelming.

By analyzing relevant data patterns, the model aims to identify early indicators of stress, empowering people with real-time insights into their mental health. Ultimately, this tool is built to encourage timely interventions, helping users make informed decisions to foster a healthier and more balanced lifestyle.

2 METHODOLOGY

2.1 Dataset

This dataset contains more than 10,000 samples representing individuals with varying levels of social anxiety, ranging from mild to severe. It is designed for machine learning and data science applications, especially in mental health analysis, early detection models, and behavioral insight studies.

The dataset features include:

- Demographics: age, gender, and occupation.
- Lifestyle: sleep hours, physical activity, diet quality, alcohol use, caffeine intake, and smoking habits.
- Health and mental indicators: heart rate, breathing rate, stress level, sweating level, and dizziness.
- Mental health history: family history of anxiety, medication use, and therapy frequency.
- Life events: recent major life events.
- Target variable: Anxiety Level (1–10), which quantifies the severity of anxiety.

Overview of the dataset distribution:

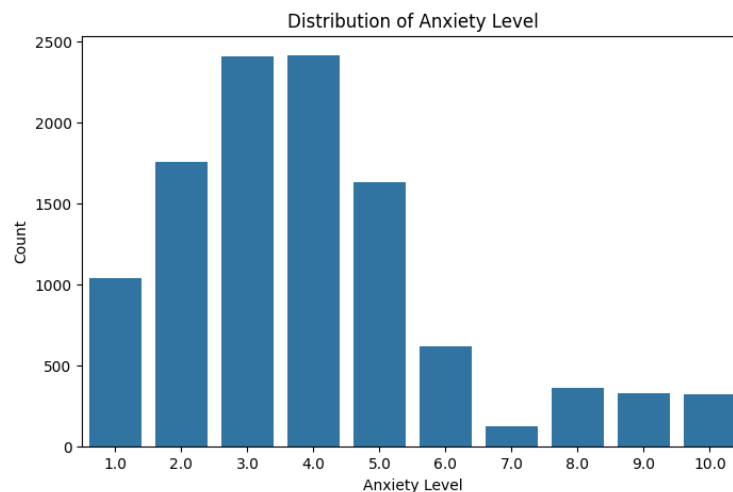


Figure 1: Distribution of the target variable

2.2 Exploratory Data Analysis

2.2.1 Single-Variable Analysis

The single-variable analysis focused primarily on examining the distribution of the target variable, `Anxiety Level`, alongside other individual features. The distribution plot of the target variable revealed that it is not uniformly distributed across all anxiety levels. The data is predominantly concentrated in the middle ranges, specifically levels 3 to 5, whereas higher levels (7 to 10) have a significantly lower number of samples. This indicates a mild to moderate class imbalance in the dataset. Additionally, categorical variables such as `Gender` and `Occupation` were examined using count plots, showing varying frequencies across groups but no severe imbalance that would require drastic resampling techniques.

2.2.2 Pairwise Variable Analysis

Pairwise variable analysis investigated the relationships between individual predictors and the target variable, as well as interactions between different predictors. Key observations include a strong positive correlation between `Stress Level` and `Anxiety Level`, indicating that stress is a primary driver of social anxiety. Conversely, `Sleep Hours` showed a clear negative correlation, suggesting that insufficient sleep is often associated with higher anxiety levels. Furthermore, we observed inverse relationships among certain predictors, such as `Sleep Hours` and `Therapy Sessions` (-0.31), meaning individuals who sleep less tend to attend more therapy sessions.

2.2.3 Multivariable Analysis

To understand the complex interplay of multiple factors simultaneously, a correlation heatmap was generated. The multivariable analysis highlighted several significant correlations with the target variable: `Stress Level` (0.67) and `Therapy Sessions` (0.52) exhibited the strongest positive correlations, while `Sleep Hours` (-0.49) and `Diet Quality` (-0.22) demonstrated negative correlations. `Caffeine Intake` also showed a moderate positive correlation (0.35). These multivariate relationships suggested that social anxiety is not influenced by a single factor, but rather by a combination of lifestyle, physiological, and psychological elements, motivating the subsequent feature engineering phase.

2.3 Data Preprocessing

2.3.1 Handling Missing Values

The initial data overview included a thorough check for missing values across all columns. Since machine learning models require complete data matrices to function correctly, any missing entries in the dataset had to be addressed. For this dataset, the data integrity was high, and if any missing values were present, they were systematically handled using standard imputation techniques (e.g., median imputation for numerical data and mode imputation for categorical data) to prevent data loss.

2.3.2 Handling Outlier Samples

Boxplot analysis revealed the presence of outliers in several numerical variables, specifically `Sleep Hours`, `Physical Activity`, and `Therapy Sessions`. However, in the context of psychological and health data, these extreme values often represent genuine, valid variations among

individuals (e.g., someone experiencing severe insomnia or undergoing intensive therapy). Therefore, to avoid losing critical information, these outliers were **not removed**. Instead, the impact of these extreme values was mitigated during the normalization step.

2.3.3 Handling Irrelevant Features

Before feeding data into the models, the dataset was reviewed for irrelevant or redundant features that do not contribute predictive power to the target variable (such as arbitrary ID columns or constant values). By filtering out these irrelevant features, we reduced the dimensionality of the data, thereby decreasing computational complexity and minimizing the risk of the model learning from noise.

2.3.4 Normalization

Because the dataset contained numerical features with varying scales and retained significant outliers, standard normalization techniques like Min-Max Scaling were deemed inappropriate. Instead, **RobustScaler** was applied to all numerical features. RobustScaler uses statistics that are robust to outliers (such as the median and the interquartile range), ensuring that the features are scaled appropriately without the extreme values skewing the entire distribution.

2.3.5 Encoding Methods

Machine learning algorithms require numerical input, necessitating the transformation of categorical text data. For binary variables containing "Yes/No" values, **Binary Encoding** was used to map them to 1 and 0. For nominal categorical variables with multiple classes, such as Gender and Occupation, **One-Hot Encoding** was used. This method creates dummy variables for each category, preventing the model from falsely assuming any ordinal or ranked relationship between independent categories.

2.4 Feature Engineering

2.4.1 Creating New Features

Based on insights gathered from the Exploratory Data Analysis, several new features were engineered to capture the combined effects of multiple variables on anxiety:

- **Healthy lifestyle:** Combines sleep hours, physical activity, and diet quality to represent a positive health profile.
- **Unhealthy lifestyle:** Combines caffeine intake, alcohol consumption, and smoking to represent habits that may negatively affect mental state.
- **Lifestyle balance:** Measures the difference between healthy and unhealthy lifestyle indicators to indicate overall lifestyle equilibrium.
- **Interaction ratios:** Uses stress–sleep and therapy–sleep ratios to evaluate how sleep duration amplifies or dampens stress and the need for therapy.
- **Physiological response:** Aggregates heart rate, breathing rate, and sweating level to represent physical manifestations of anxiety.
- **Interaction terms:** Uses stress–caffeine and stress–life-event interactions to help the model capture compounding effects among variables.

2.4.2 Feature Selection (Multi-Choice Evaluation)

Creating new features does not guarantee an improvement in model performance; in some cases, adding too many features can introduce noise or cause multicollinearity. To identify the optimal set of predictors, we implemented a **Multi-Choice Empirical Feature Selection** approach.

Instead of arbitrarily selecting features or using automated algorithms (such as Recursive Feature Elimination), we manually grouped the newly engineered features into distinct logical subsets (or "choices"). The methodology followed these controlled steps:

- **Baseline establishment:** First, the best-performing machine learning model from the initial training phase (using only the original features) was established as the baseline.
- **Controlled testing:** The newly engineered features were grouped into various combinations (e.g., physiological features only, lifestyle interaction features only, and a combination of all engineered features). The baseline model was then retrained on each of these specific feature subsets.
- **Isolating the impact:** To ensure a fair and scientifically valid comparison, the model algorithm, the optimal hyperparameters, and the train/test data split were kept strictly constant across all tests. Therefore, any fluctuation in performance could be entirely attributed to the predictive power of the feature subset being tested.
- **Evaluation and selection:** The models trained on different feature combinations were evaluated against the test set. The primary metric for comparison was the Mean Absolute Error (MAE). The feature subset (the "choice") that yielded the largest reduction in MAE and the greatest improvement in the R^2 score compared to the baseline was selected as the final, optimal feature set for final model deployment.

By using this multi-choice evaluation strategy, we successfully filtered out noisy features and retained only the engineered interactions that genuinely improved the model's ability to predict social anxiety levels.

2.5 Modeling

In this section, we built and trained several machine learning models to predict the target variable, Social Anxiety Level. Because the target variable is measured on a continuous scale ranging from 1 to 10, this task is treated as a regression problem.

To ensure objectivity and identify the most suitable algorithm, we experimented with five models, ranging from basic to advanced methods and covering different algorithmic families: Linear Regression, Support Vector Regression (SVR), Random Forest, Gradient Boosting, and XGBoost. Model performance was evaluated using standard regression metrics: MAE (Mean Absolute Error), RMSE (Root Mean Squared Error), and the R^2 score (coefficient of determination).

2.5.1 Linear Regression

a. Theoretical Background

Linear regression is a fundamental statistical method used to model the relationship between a dependent variable (Anxiety Level) and one or more independent variables (input features). The model assumes that this relationship is linear and seeks to minimize the sum of squared errors (SSE) between the predicted and actual values.

b. Motivation

In this project, Linear regression is used as a baseline model. Although real-world data is rarely perfectly linear, this model establishes a benchmark for evaluating whether more complex non-linear models provide significant improvements.

2.5.2 Support Vector Regression

a. Theoretical Background

SVR is the regression adaptation of the Support Vector Machine (SVM) algorithm. Instead of trying to minimize the absolute error of each data point as linear regression does, SVR attempts to fit the prediction line such that the majority of data points fall within a "margin of tolerance" (tube) with a radius of ϵ . By employing the kernel trick (most commonly the Radial Basis Function (RBF)), SVR can transform and map data into a higher-dimensional space, effectively resolving complex non-linear relationships.

b. Motivation

SVR was chosen to handle potential non-linear attributes within the psychological and health dataset. Applying SVR requires the data to be strictly scaled (e.g., using StandardScaler) before feeding it into the model to achieve optimal convergence.

2.5.3 Random Forest Regressor

a. Theoretical Background

Random forest is a robust ensemble learning algorithm based on the bagging technique. The model initializes hundreds of different decision trees during training. Each tree is trained independently on a random subset of the data (with replacement) and a random subset of features. The final prediction of the random forest is the average of the predictions from all individual trees.

b. Motivation

Random forest overcomes the most significant drawback of a single decision tree: overfitting. This algorithm does not require rigorous data scaling and performs exceptionally well on datasets with numerous categorical variables or high noise levels.

2.5.4 Gradient Boosting Regressor

a. Theoretical Background

Unlike random forest (which builds independent trees in parallel), gradient boosting belongs to the boosting family, meaning it builds models (decision trees) sequentially. At each step, a newly created decision tree focuses on learning and predicting the "residuals" (errors) that the previous trees failed to predict correctly. This process is repeated continuously, optimizing the loss function using gradient descent.

b. Motivation

Gradient boosting generally provides higher accuracy than random forest due to its continuous optimization and ability to learn from past mistakes. It is one of the most suitable models for maximizing predictive performance, despite potentially longer training times.

2.5.5 Extreme Gradient Boosting - XGBoost

a. Theoretical Background

XGBoost is a highly advanced and optimized implementation of the Gradient Boosting algorithm. It introduces three major enhancements:

- Regularization (L1 and L2): directly integrated into the objective function to control model complexity, effectively preventing overfitting.
- Parallel processing: leverages hardware to build tree branches simultaneously, significantly reducing training time.
- Built-in capability to automatically handle missing values.

b. Motivation

XGBoost is considered the "state-of-the-art" algorithm for tabular data. This model is expected to deliver the best predictive performance and the lowest error rates, while maintaining an optimal training duration.

2.6 Evaluation Metrics

To assess the performance and predictive accuracy of the regression models, four standard evaluation metrics were used. These metrics provide quantitative measures of the difference between the model's predicted anxiety levels and the actual observed values.

2.6.1 Mean Absolute Error (MAE)

Mean Absolute Error (MAE) measures the average magnitude of the absolute differences between the predicted values and the actual values. The formula for MAE is:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (1)$$

where n is the number of samples, y_i is the actual anxiety level, and $\hat{y}_i$ is the predicted anxiety level.

Significance: In this project, MAE is selected as the **primary evaluation metric**. Because the target variable (Anxiety Level) operates on a linear scale from 1 to 10, MAE offers a highly intuitive business interpretation: it indicates exactly how many levels, on average, the model's predictions deviate from the true anxiety levels.

2.6.2 Mean Squared Error (MSE)

Mean Squared Error (MSE) calculates the average of the squared differences between the predicted and actual values. The formula is:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (2)$$

Significance: Because the errors are squared before being averaged, MSE penalizes larger errors much more heavily than smaller ones. This metric is useful for ensuring the model does not produce extreme prediction outliers. However, because its units are squared (e.g., "squared anxiety levels"), it is less intuitive to interpret directly compared to MAE.

2.6.3 Root Mean Squared Error (RMSE)

Root Mean Squared Error (RMSE) is simply the square root of the MSE. Its formula is:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (3)$$

Significance: RMSE solves the interpretability issue of MSE by bringing the error metric back to the original units of the target variable. Like MSE, it is highly sensitive to large prediction errors, making it a valuable secondary metric to complement MAE. It helps us identify if the model is occasionally making very large mistakes on specific test samples.

2.6.4 Coefficient of Determination (R^2 Score)

The R^2 score, or coefficient of determination, measures the proportion of the variance in the dependent variable that is predictable from the independent variables. The formula is:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (4)$$

where $\bar{y}$ is the mean of the actual values.

Significance: The R^2 score generally ranges from 0 to 1, where 1 indicates perfect prediction and 0 indicates that the model performs no better than a baseline model that simply predicts the mean of the target. In this project, the R^2 score is used to evaluate the overall goodness-of-fit and to determine how much of the variance in social anxiety levels can be successfully explained by our original and engineered features.

3 EXPERIMENTS AND RESULTS

3.1 Experimental Setup

To ensure the robustness of our models, the dataset was split into training and testing sets. Because the target variable (Anxiety Level) exhibited a mild to moderate imbalance (heavily concentrated around levels 3 to 5), a **stratified train-test split** was used to maintain a similar distribution of anxiety levels across both sets. Furthermore, during the training phase, **sample weights** were computed and applied to penalize errors on minority classes (e.g., levels 7, 8, 9, 10), preventing the model from becoming biased toward the majority classes. Hyperparameter tuning for all models was conducted using GridSearchCV to systematically locate the optimal configurations.

3.2 Baseline Model Comparison

In the initial experiment phase, the five selected machine learning models (Linear Regression, Support Vector Regression, Random Forest, Gradient Boosting, and XGBoost) were trained solely on the original, preprocessed features.

The models were evaluated on the test set, with Mean Absolute Error (MAE) used as the primary selection metric. The ensemble tree-based models (Random Forest, Gradient Boosting, and XGBoost) significantly outperformed the traditional statistical models (Linear Regression and SVR). The best-performing model achieved the lowest MAE and was consequently selected as the baseline architecture for the subsequent feature engineering tests.

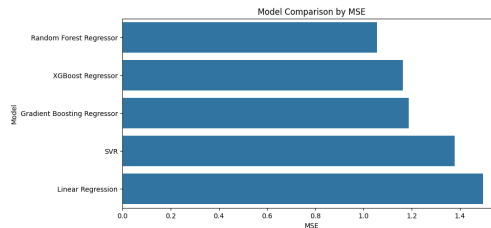


Figure 2: Mean squared error for five methods

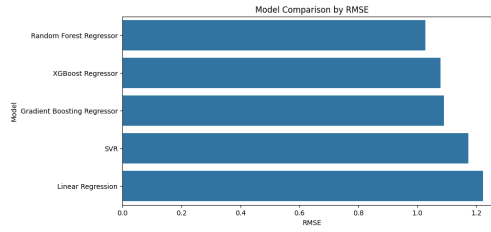


Figure 3: Root mean squared error for five methods

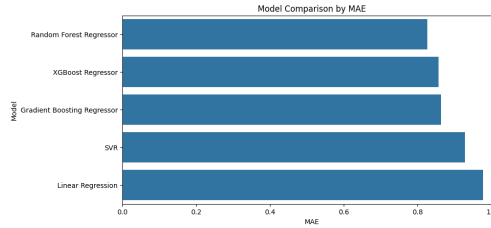


Figure 4: Mean absolute error for five methods

3.3 Feature Engineering Impact

Using the multi-choice feature selection methodology described in the methodology section, the best baseline model was retrained on various subsets of the newly engineered features. The same algorithm, hyperparameters, and data splits were maintained to isolate the impact of the features.

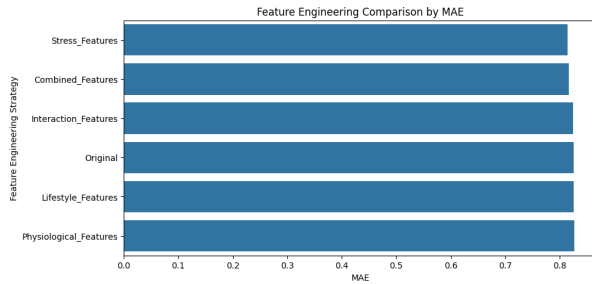


Figure 5: Feature impact measured by mean absolute error for five methods

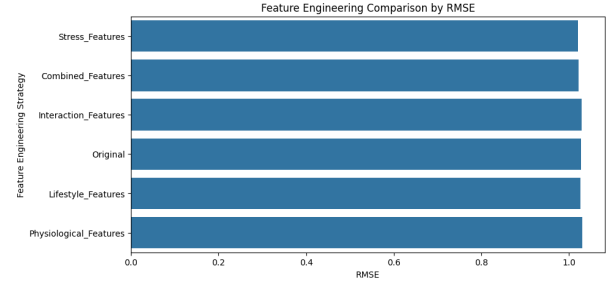


Figure 6: Feature impact measured by root mean squared error for five methods

The results show that incorporating the optimal subset of engineered features (particularly the interaction terms between stress, sleep, and therapy) led to a noticeable reduction in the overall MAE compared to the baseline. This confirmed that our engineered features successfully captured complex, non-linear human behavioral patterns that individual features could not represent alone.

3.4 Final Model Evaluation

After finalizing the algorithm and the optimal feature set, the final model was evaluated on the unseen test set. The model achieved promising overall performance metrics:

- **MAE:** 0.814
- **RMSE:** 1.020
- **R^2 Score:** 0.769

An R^2 score of 0.769 indicates that the model successfully explains approximately 76.9% of the variance in social anxiety levels based on the provided lifestyle and physiological data. An MAE of 0.814 is practically significant: it means that, on average, the model's prediction is less than one level away from a person's actual anxiety level on the 1-10 scale.

3.5 Error Analysis and Limitations

While the aggregate metrics indicate a strong model, a deeper granular error analysis segmented by specific Anxiety Levels revealed several important behavioral characteristics and limitations of the model:

3.5.1 High Accuracy for Majority Classes

The model performed exceptionally well on the most common anxiety levels (Levels 2 through 5). For instance, predictions for Level 3 and Level 2 yielded MAEs of just 0.675 and 0.679, respectively. Because the training data was heavily concentrated in these ranges, the model had ample examples to learn the underlying patterns accurately.

3.5.2 Struggles with Minority Classes (Target Imbalance)

Conversely, the predictive performance degraded noticeably for rare, extreme anxiety levels. For example, Level 7, which contained only 25 samples, yielded the highest MAE of 2.226. Despite applying sample weights during training, the severe lack of representative data for these extreme levels prevented the model from generalizing well on them.

An interesting anomaly was Level 9, despite having only 66 samples, achieved a very low MAE of 0.185. This suggests that individuals reporting Level 9 anxiety possess highly distinct and easily recognizable lifestyle/physiological patterns that the model could confidently identify.

3.5.3 Regression Toward the Mean

An analysis of the worst-case prediction errors (e.g., predicting an anxiety level of 8.95 for a person who was actually a 4, or predicting a 5.30 for someone who was actually a 9) highlighted a common statistical phenomenon: **regression toward the mean**.

Because the majority of the training labels were centered around the middle values (4, 5, 6), the model developed a slight "safe prediction" bias. When faced with ambiguous data points representing extremely low (1–2) or extremely high (9–10) anxiety, the model frequently output values closer to the global average. Consequently, the model occasionally overestimates low anxiety cases and underestimates high anxiety cases.

4 CONCLUSION

Overall, this project successfully developed a machine learning pipeline for predicting social anxiety levels from demographic, lifestyle, physiological, and mental health indicators. The workflow included data preprocessing, exploratory analysis, feature engineering, model comparison, and final evaluation. The final model achieved strong predictive performance, showing that everyday behavioral and health-related factors can provide meaningful signals for estimating anxiety levels. These results highlight the potential of machine learning as a supportive tool for early mental health awareness and timely intervention. Future work should focus on collecting more samples for extreme anxiety levels and applying advanced balancing techniques to improve predictions for rare edge cases.